6G4Z3006 Calculus
Department of Computing and Mathematics
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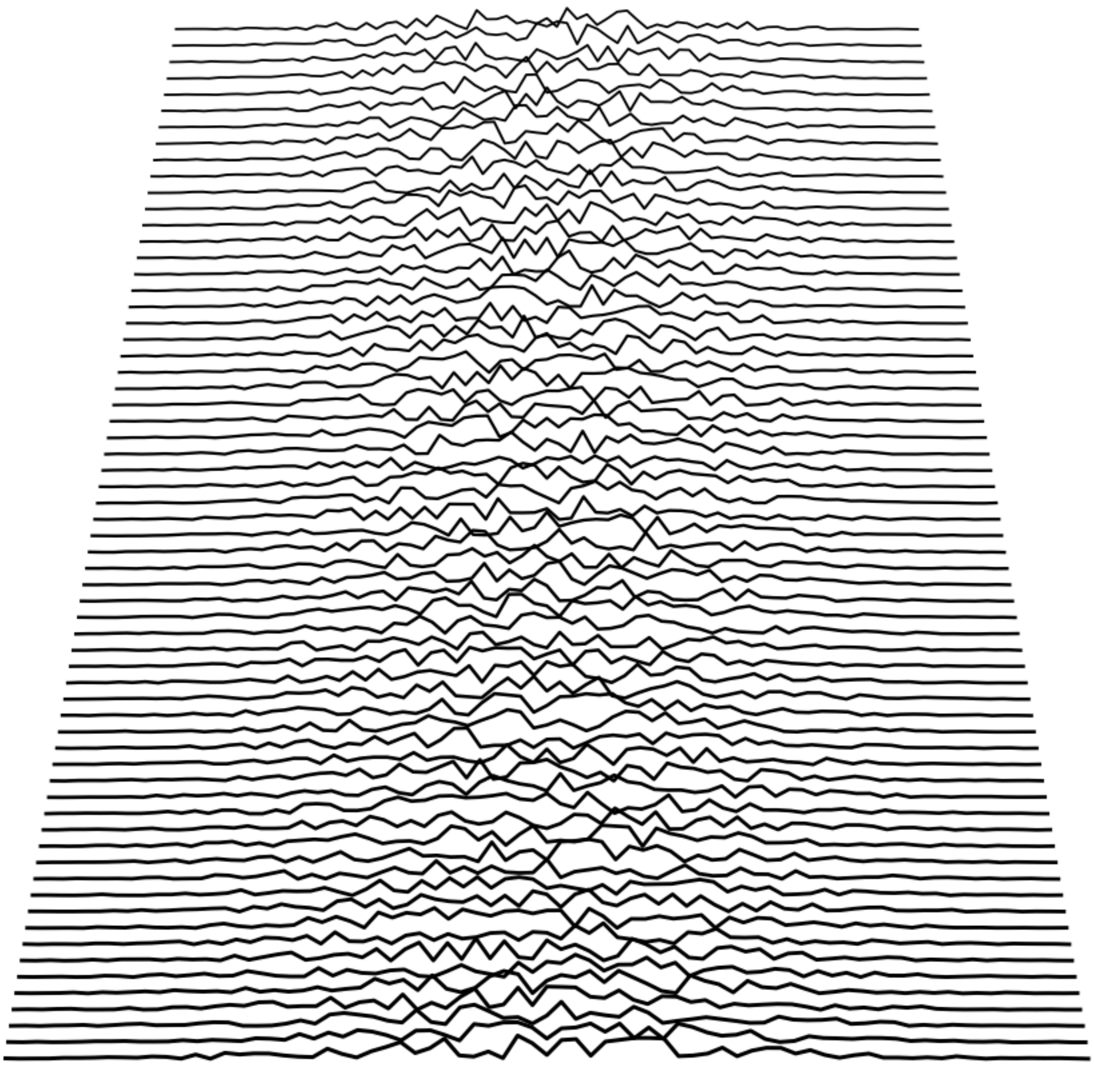
Manchester Metropolitan University

Calculus

The Solutions

Worked solutions to most of the exercises in
Sequences, Series & Differentiation



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Sequences Exercises 1.5

1. a) $1, 2, 4, 8, 16, \ldots$
   b) $2, \frac{4}{3}, \frac{6}{5}, \frac{8}{7}, \frac{10}{9}, \ldots$
   c) $-1, 2, -\frac{3}{2}, \frac{2}{3}, -\frac{5}{24}, \ldots$

2. a) $x_{2m+3} = 2^{2m+2}$,
   b) $x_{2m+3} = \frac{4m+6}{4m+5}$,
   c) $x_n = -\frac{(2m+3)^2}{(2m+3)!}$,
   d) $x_n = \cos\frac{(2m+3)\pi}{4}$.

3. a) The difference between succesive terms is

$$\begin{aligned} x_{n+1} - x_n &= \frac{n+1}{2n+3} - \frac{n}{2n+1} \\ &= \frac{(n+1)(2n+1) - n(2n+3)}{(2n+3)(2n+1)} \\ &= \frac{1}{(2n+3)(2n+1)} > 0, \text{ for all } n \geq 1, \end{aligned}$$

   and therefore $\{x_n\}$ is an increasing sequence.

   b) The ratio of successive terms is

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{10^{n+1}}{(2n+2)!}\frac{(2n)!}{10^n} \\ &= \frac{10}{(2n+2)(2n+1)} \\ &= \frac{10}{4n^2+6n+2} < 1, \text{ for all } n \geq 1. \end{aligned}$$

   Hence $x_{n+1} < x_n$ for all $n \geq 1$ and therefore $\{x_n\}$ is a decreasing sequence.

   c) The ratio of successive terms is

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{e^{n+1}}{(n+2)!}\frac{(n+1)!}{e^n} \\ &= \frac{e}{n+2} \\ &< 1, \text{ for all } n \geq 1 \text{ since } e < 3. \end{aligned}$$

   Hence $x_{n+1} < x_n$ for all $n \geq 1$ and therefore $\{x_n\}$ is a decreasing sequence.

4. a) Apply l'Hôpital's rule , $x_n \to 0$,
   b) Apply l'Hôpital's rule , $x_n \to 0$,
   c) Divide numerator and denominator by n^2 and then apply agebra of limits theorem, $x_n \to \frac{1}{2}$,
   d) Divergent. x_n increases without upper bound. Show by establishing that $x_n > 2n$.
   e) Apply algebra of limits theorem, $x_n \to 1$.

5. $\{r_n\}$ is strictly decreasing as $r_n = \frac{2n+1}{2n+2}r_{n-1}$ for all $n \geq 2$. Also $\{r_n\}$ is bounded below by 0 (since all factors in r_n are positive). So by the Monotone Convergence theorem $\{r_n\}$ is convergent.

6. Note that $a_4 = \frac{24}{16} > 1$. Also, for $m \geq 1$ we have

$$\begin{aligned} a_{4+m} &= \frac{4+m}{2} a_{3+m} \\ &> 2\, a_{3+m}. \end{aligned}$$

Putting these facts together allows us to establish that

$$a_{4+m} > 2^m,$$

which clearly shows that the sequence $\{a_n\}$ is unbounded and hence divergent by theorem 1.2 .

Really this argument should be given as a 'proof by induction'. We shall return to it and construct it as such a proof after you have been introduced to the proof by induction method later in the course.

7. Let $\{x_n\}$ be the arithmetic sequence where $x_n = a + (n-1)d$. The partial sum S_m is given by

$$S_m = \sum_{n=1}^{m} (a + (n-1)d) = a + (a+d) + (a+2d) + \cdots + (a + (m-2)d) + (a + (m-1)d).$$

Following the hint we consider the quantity $2S_m$ where S_m is added to itself in reverse order. In this sum the term $(a + (i-1)d)$ will be added to $(a + (m-i)d)$ to give $2a + (m-1)d$, for each $i = 1, \ldots, m$. Therefore $2S_m = m(2a + (m-1)d)$ and so the required formula is

$$S_m = ma + \frac{m}{2}(m-1)d.$$

8. Not included.

9. a)

$$\begin{aligned} a_n &= \frac{(2n+1)(n-2)}{n^3+2} \\ &= \frac{2n^2 - 3n - 2}{n^3 + 2} \\ &= \frac{\frac{2}{n} + \frac{-3}{n^2} + \frac{-2}{n^3}}{1 + \frac{2}{n^3}} \end{aligned}$$

Now each of the terms in the numerator represent sequences that converge to the limit 0 and so by part (1) of algebra of sequences theorem pg. 62, the numerator converges to 0.

Applying part (1) of algebra of sequences theorem to the denominator, where 1 represents the constant sequence of 1s, we see that the denominator tends to 1. Hence, by part (3) of the theorem the sequence $\{a_n\}$ converges to 0.

We can express this in symbols as

$$\begin{aligned} \lim_{n\to\infty} a_n &= \lim_{n\to\infty} \left(\frac{\frac{2}{n} + \frac{-3}{n^2} + \frac{-2}{n^3}}{1 + \frac{2}{n^3}} \right) \\ &= \frac{\lim_{n\to\infty} \left(\frac{2}{n} + \frac{-3}{n^2} + \frac{-2}{n^3} \right)}{\lim_{n\to\infty} \left(1 + \frac{2}{n^3} \right)} \\ &= \frac{0}{1} \end{aligned}$$

b)

$$\begin{aligned}\lim_{n\to\infty} b_n &= \lim_{n\to\infty}\left(\frac{2}{7n^2} - \frac{3n}{n^2} + \frac{2n}{7n+1}\right)\\ &= \lim_{n\to\infty}\left(\frac{2}{7n^2}\right) - \lim_{n\to\infty}\left(\frac{3}{n}\right) + \lim_{n\to\infty}\left(\frac{2}{7+\frac{1}{n}}\right)\\ &= 0 - 0 + \frac{2}{7}\end{aligned}$$

10. Suppose that the sequence $\{c_n\}$ does converge to a non zero limit l.

    Point out that the sequences $\{c_n\}$ and $\{c_{n+1}\}$ are the same sequence really and so $\lim_{n\to\infty} c_n = \lim_{n\to\infty} c_{n+1} = l$.

    Applying the algebra of sequences theorem to the recurrence relation give

$$\begin{aligned}l &= \lim_{n\to\infty} c_{n+1}\\ &= \lim_{n\to\infty}\left(\frac{2}{3}c_n + \frac{2}{3c_n^2}\right)\\ &= \frac{2}{3}l + \frac{2}{3l^2}\end{aligned}$$

    Multiplying across by l^2 gives

$$\begin{aligned}l^3 &= \frac{2}{3}l^3 + \frac{2}{3}\\ \Rightarrow \frac{1}{3}l^3 &= \frac{2}{3}\\ \Rightarrow l^3 &= 2\\ \Rightarrow l &= 2^{\frac{1}{3}}.\end{aligned}$$

11. Not included.

Series Exercises 2.5

1. a) This series is a sum of terms, each of which is greater than 1, so it is clearly divergent. We can prove it explicitly using the formula for the sum of the first k terms of an arithmetic sequence i.e.,

$$\sum_{n=1}^{k} a + (n-1)d = ka + \frac{k}{2}(k-1)d.$$

This gives us

$$\begin{aligned} S_k = \sum_{n=1}^{k}(2n-1) &= \sum_{n=1}^{k}(1+(n-1)2), \text{ getting it into standard form} \\ &= k + \frac{k}{2}(k-1)2 \\ &= k + k^2 - k = k^2\,. \end{aligned}$$

And so we see clearly that $\{S_k\}$ is a divergent sequence, and hence the original series diverges.

b) This series is the sum of a geometric sequence, with initial term 1 and common ratio $\frac{1}{2}$. The formula for the sum of the first k terms of a geometric sequence is

$$\sum_{n=1}^{k} ar^{n-1} = \frac{a(1-r^k)}{1-r}.$$

This gives us

$$\begin{aligned} S_k &= \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n-1} \\ &= \frac{1-\left(\frac{1}{2}\right)^k}{\frac{1}{2}} \\ &= 2 - \left(\tfrac{1}{2}\right)^{k-1} \end{aligned}$$

This shows that $\{S_k\}$ is a sum of two convergent sequences and

$$\begin{aligned} \lim_{k\to\infty} S_k &= \lim_{k\to\infty}(2 - \left(\tfrac{1}{2}\right)^{k-1}) \\ &= 2 - 0 = 2\,. \end{aligned}$$

c) We use the method of partial fractions to rewrite the fractio. Partial fractions will be covered in detail later in the unit. Here we see an example of how they can be used.

$$\frac{1}{n(n+1)} = \frac{\alpha}{n} + \frac{\beta}{n+1},$$

where α and β are constants to be determined. Multiplying both sides by $n(n+1)$ we get

$$\begin{aligned} 1 &= \alpha(n+1) + \beta n \\ &= (\alpha+\beta)n + \alpha. \end{aligned}$$

This can only be true for all n if $\alpha = 1$ and $\beta = -1$. So

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Now we see that when we consider the partial sums, S_k, of the original series there will be cancellation between succesive terms.

$$\begin{aligned}S_k &= \sum_{n=1}^{k} \frac{1}{n(n+1)} \\ &= \sum_{n=1}^{k} \frac{1}{n} - \frac{1}{n+1} \\ &= 1 - \frac{1}{k+1}.\end{aligned}$$

This shows that $\{S_k\}$ is a sum of two convergent sequences and

$$\begin{aligned}\lim_{k\to\infty} S_k &= \lim_{k\to\infty} (1 - \frac{1}{k+1}) \\ &= 1 - 0 = 1\end{aligned}$$

So this is a convergent series and

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

2. a) The sequence of partial sums here is the alternating sequence $-1, 0, -1, 0, -1, 0 \ldots$ which is clearly divergent.

   b)
$$\sum_{n=1}^{\infty} (-1)^n \left(\frac{n+1}{n+2}\right) = \frac{-2}{3} + \frac{3}{4} - \frac{4}{5} + \frac{5}{6} - \ldots$$

   Let $\{S_k\}$ denote the sequence of partial sums of this series. Notice that each term in the series has greater magnitude than the previous term and so $\{S_k\}$ begins

$$\begin{aligned}S_1 &= \frac{-2}{3} < 0 \\ S_2 &= \frac{-2}{3} + \frac{3}{4} > 0 \\ S_3 &= \frac{-2}{3} + \frac{3}{4} - \frac{4}{5} < S_1 < 0 \\ S_4 &= \frac{-2}{3} + \frac{3}{4} - \frac{4}{5} + \frac{5}{6} > S_2 > 0 \\ &\vdots\end{aligned}$$

   So we see that $\{S_k\}$ is an alternating sequence with the negative terms getting larger in absolute value, i.e. moving away from 0, and the positive terms getting larger in absolute value, i.e. moving away from 0.
   This shows that $\{S_k\}$ is a divergent sequence and so the original series is divergent also.

3. Not included.

4. a)
$$\begin{aligned}x_n &= \tfrac{2}{3}\left(\tfrac{1}{2}\right)^n \tfrac{1}{4}\left(\tfrac{2}{3}\right)^n \\ &= \tfrac{2}{3}\tfrac{1}{4}\left(\tfrac{1}{2}\tfrac{2}{3}\right)^n \\ &= \tfrac{1}{6}\left(\tfrac{1}{3}\right)^n \\ &= \tfrac{1}{18}\left(\tfrac{1}{3}\right)^{n-1}\end{aligned}$$

So we see that the series is a geometric series with initial term $\frac{1}{18}$ and common ration $\frac{1}{3} < 1$. Therefore the series is convergent and the limit is

$$\sum_{n=1}^{\infty} x_n = \frac{\frac{1}{18}}{1 - \frac{1}{3}} = \frac{1}{12}.$$

b)

$$\begin{aligned} x_n &= \left(\tfrac{3}{2}\right)^{2n} \tfrac{1}{2} \left(\tfrac{2}{3}\right)^{n+1} \\ &= \tfrac{1}{2} \left(\tfrac{3}{2}\right)^{2n} \left(\tfrac{3}{2}\right)^{-n-1} \\ &= \tfrac{1}{2} \left(\tfrac{3}{2}\right)^{n-1}. \end{aligned}$$

So we see that x_n is a geometric series with initial term $\frac{1}{2}$ and common ratio $\frac{3}{2}$. However the common ratio $\frac{3}{2} > 1$, and so this series is divergent.

5. To find the partial fraction expansion of $\frac{2}{n^2+2n}$ we first factorize the denominator as

$$x_n = \tfrac{2}{n^2+2n} = \tfrac{2}{n(n+2)}$$

and then seek an expansion for this of the form

$$\tfrac{2}{n(n+2)} = \tfrac{\alpha}{n} + \tfrac{\beta}{n+2}.$$

Expressing the right hand side as a single fraction gives

$$\tfrac{2}{n(n+2)} = \tfrac{\alpha(n+2)+\beta n}{n(n+2)}$$

. The only way that $\alpha(n+2) + \beta n = 2$ is if $\alpha = 1$ and $\beta = -1$. Ans so our infinite series can be written as

$$\sum_{n=1}^{\infty} \frac{2}{n^2 + 2n} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right).$$

Therefore in the partial sum S_k we will get some cancellation between terms leaving S_k as

$$S_k = \sum_{n=1}^{k} \left(\frac{1}{n} - \frac{1}{n+2} \right) = 1 + \frac{1}{2} - \frac{1}{k+1} - \frac{1}{k+2}.$$

Hence the sequence of partial sums $\{S_k\}$ is convergent and

$$\lim_{k\to\infty} S_k = \lim_{k\to\infty} \left(1 + \frac{1}{2} - \frac{1}{k+1} - \frac{1}{k+2} \right) = \frac{3}{2}.$$

Therefore the infinite series is convergent and

$$\sum_{n=1}^{\infty} x_n = \frac{3}{2}.$$

6. We can express the exapansion as

$$\begin{aligned} 2.104104104104\ldots &= 2 + \tfrac{104}{10^3} + \tfrac{104}{10^6} + \tfrac{104}{10^9} + \ldots \\ &= 2 + \tfrac{104}{10^3} \left(1 + \tfrac{1}{10^3} + \tfrac{1}{10^6} + \ldots\right) \\ &= 2 + \tfrac{104}{10^3} \left(1 + \tfrac{1}{10^3} + \left(\tfrac{1}{10^3}\right)^2 + \ldots\right) \end{aligned}$$

The part in brackets in this last expression we recognise as a geometric series with initial term 1 and common ratio $\frac{1}{10^3}$ and so we get

$$\begin{aligned}2.104104104104\ldots &= 2 + \tfrac{104}{10^3}\left(\tfrac{1}{1-\frac{1}{10^3}}\right)\\ &= 2 + \tfrac{104}{10^3}\left(\tfrac{10^3}{999}\right)\\ &= 2 + \tfrac{104}{999}\end{aligned}$$

7. We will show that the series $\sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)}$ diverges by using the comparison test and the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$.

$$\begin{aligned}\frac{2n+1}{n(n+1)} &= \frac{2n+1}{n^2+n)}\\ &= \frac{\frac{2}{n}+\frac{1}{n^2}}{1+\frac{1}{n}}\\ &> \frac{\frac{2}{n}}{1+1}, \text{ (we have made the num. smaller and the denom. bigger)}\\ &= \frac{1}{n}\end{aligned}$$

So each term of the original series is greater than the corresponding term of the divergent harmonmic series. Therefore $\sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)}$ diverges.

8. a) We will use the ratio test on the series $\sum_{n=1}^{\infty} a_n$ where $a_n = \frac{2}{n!}$.

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{2}{(n+1)!}\frac{n!}{2}\\ &= \frac{1}{n+1}\end{aligned}$$

Therefore

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to\infty}\frac{1}{n+1} = 0$$

and hence by the ratio test $\sum_{n=1}^{\infty} \frac{2}{n!}$ is a convergent series.

b) We will use the ratio test on the series $\sum_{n=1}^{\infty} b_n$ where $b_n = \frac{n!}{n^n}$.

$$\begin{aligned}\frac{b_{n+1}}{b_n} &= \frac{(n+1)!}{(n+1)^{n+1}}\frac{n^n}{n!}\\ &= \frac{(n+1)n^n}{(n+1)^{n+1}}\\ &= \frac{n^n}{(n+1)^n}\\ &= \left(\frac{n}{n+1}\right)^n\\ &= \left(\frac{1}{1+\frac{1}{n}}\right)^n\\ &= \left(1+\frac{1}{n}\right)^{-n}\end{aligned}$$

Using the well known result that

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e,$$

we see that

$$\lim_{n\to\infty}\frac{b_{n+1}}{b_n} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{-n} = e^{-1} < 1.$$

Therefore the series $\sum_{n=1}^{\infty}\frac{n!}{n^n}$ is a convergent series.

c) We will use the root test on the series $\sum_{n=1}^{\infty} c_n$, where $c_n = \left(\frac{2n-1}{3+4n}\right)^n$.

$$\begin{aligned}(c_n)^{\frac{1}{n}} &= \frac{2n-1}{3+4n}\\ &= \frac{2-\frac{1}{n}}{4+\frac{3}{n}}\end{aligned}$$

Therefore

$$\lim_{n\to\infty}(c_n)^{\frac{1}{n}} = \lim_{n\to\infty}\frac{2-\frac{1}{n}}{4+\frac{3}{n}} = \frac{1}{2},$$

and so by the root test the series $\sum_{n=1}^{\infty}\left(\frac{2n-1}{3+4n}\right)^n$ converges.

d) We will use the ratio test on the series $\sum_{n=1}^{\infty} d_n$, where $d_n = \frac{(n!)^2}{(2n)!}$.

$$\begin{aligned}\frac{d_{n+1}}{d_n} &= \frac{(n+1)!^2}{(2n+2)!}\frac{(2n)!}{n!^2}\\ &= \frac{(n+1)^2}{(2n+2)(2n+1)}\\ &= \frac{n^2+2n+1}{4n^2+6n+2}\\ &= \frac{1+\frac{2}{n}+\frac{1}{n^2}}{4+\frac{6}{n}+\frac{2}{n^2}}\end{aligned}$$

Therefore

$$\lim_{n\to\infty}\frac{d_{n+1}}{d_n} = \lim_{n\to\infty}\frac{1+\frac{2}{n}+\frac{1}{n^2}}{4+\frac{6}{n}+\frac{2}{n^2}} = \frac{1}{4} < 1,$$

and hence the series $\sum_{n=1}^{\infty}\frac{(n!)^2}{(2n)!}$ is a convergent series.

9. Use standard comparison test. Observe that

$$\frac{2n^2+4}{3n^3-2n} > \frac{2n^2}{3n^3} = \frac{2}{3}\frac{1}{n},$$

and hence $\sum a_n$ diverges as $\sum\frac{1}{n}$ is divergent.

10. Square roots can be awkward to deal with in inequalities. Often better to use the limit comparison tests. Examining the ration of highest powers int he numerator and denominator suggests using

the comparison series $\sum \frac{n}{n^3} = \sum \frac{1}{n^2}$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{2n+5}{\sqrt{n^6+n}} \frac{1}{\frac{1}{n^2}} &= \lim_{n\to\infty} \frac{2n^3+5n^2}{\sqrt{n^6+n}} \\
&= \lim_{n\to\infty} \frac{2+\frac{5}{n}}{\frac{1}{n^3}\sqrt{n^6+n}} \\
&= \lim_{n\to\infty} \frac{2+\frac{5}{n}}{\sqrt{1+\frac{1}{n^5}}} \\
&= 2
\end{aligned}$$

So since the limit of the ratios exists we conclude by the limit comparison test that $\sum b_n$ converges as $\sum \frac{1}{n^2}$ converges.

11. Use the ratio test.

$$\lim_{n\to\infty} \frac{c_{n+1}}{c_n} = \lim_{n\to\infty} \frac{(n+1)!}{5^{n+3}} \frac{5^{n+2}}{n!} = \lim_{n\to\infty} \frac{n+1}{5},$$

which is clearly divergent as it increases without upper-bound. So by the ratio test the series $\sum c_n$ is divergent.

12. Use the ratio test.

$$\begin{aligned}
\lim_{n\to\infty} \frac{d_{n+1}}{d_n} &= \lim_{n\to\infty} \frac{(n+1)^2 2^{n+6}}{2^{2n+2}} \frac{2^{2n}}{n^2 2^{n+5}} \\
&= \lim_{n\to\infty} \frac{1}{2} \frac{(n+1)^2}{n^2} \\
&= \lim_{n\to\infty} \frac{1}{2} \left(1+\frac{1}{n}\right)^2 \\
&= \frac{1}{2}.
\end{aligned}$$

Since the limit is strictly less than 1 we conclude by the ratio test that $\sum d_n$ is convergent.

13. Use the standard comparison test.

$$\frac{\sin^2(n)}{n(n+3)} \leq \frac{1}{n(n+3)} < \frac{1}{n^2}.$$

$\sum \frac{1}{n^2}$ is convergent and so by the comparison test so is $\sum e_n$.

14. Use the limit comparison test with the comparison series $\sum \frac{1}{n}$.

$$\lim_{n\to\infty} \frac{\sqrt{n+1}}{n\sqrt{n+4}} \frac{1}{\frac{1}{n}} = \lim_{n\to\infty} \frac{\sqrt{n+1}}{\sqrt{n+4}} = \lim_{n\to\infty} \frac{\sqrt{1+\frac{1}{n}}}{\sqrt{1+\frac{4}{n}}} = 1.$$

The limit of the ratios is 1 and hence by the limit comparison test the series $\sum f_n$ diverges as $\sum \frac{1}{n}$ is divergent.

15. Use the ratio test.

$$\lim_{n\to\infty} \frac{(n+1)2^{n+1}}{(n+1)!}\frac{n!}{n2^n} = \lim_{n\to\infty} \frac{2}{n} = 0.$$

The limit of the ratios is 0 and so the series $\sum g_n$ converges by the ratio test.

16. Let $f(x) = e^x$. We have the result that

$$\frac{d}{dx}(e^x) = e^x,$$

and so for all $k \geq 1$,

$$f^{(k)}(0) = 1.$$

Therefore by Taylor's theorem (applied to the interval $[0, 1]$, i.e. $x = 1$),

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + e^c \frac{x^{n+1}}{(n+1)!},$$

for some $c \in (0, 1)$.
In order to get an approximation correct to four decimal places we should arrange for the remainder term to satisfy

$$\left| e^c \frac{x^{n+!}}{(n+1)!} \right| < 0.00001.$$

Now when evaluated at $x = 1$ the remainder term satisfies

$$\begin{aligned} \left| e^c \frac{x^{n+1}}{(n+1)!} \right| &= \frac{e^c}{(n+1)!} \\ &< \frac{e}{(n+1)!}, \text{ since } c \in (0,1), \\ &< \frac{3}{(n+1)!}, \text{ assuming that } e < 3. \end{aligned}$$

Experimentation shows that $\frac{3}{9!} < 0.00001$, (i.e. taking $n = 8$) and so taking the first 9 terms of the Taylor expansion will produce an approximation for e of the desired accuracy, i.e.

$$e \approx 1 + \sum_{k=1}^{8} \frac{1}{k!} = 2.71827\ldots$$

17. We need that Taylor expansion for $\sin x$ about 0. Let $f(x) = \sin x$ and so

$$\begin{aligned} f(x) &= \sin x \\ f^{(1)}(x) &= \cos x \\ f^{(2)}(x) &= -\sin x \\ f^{(3)}(x) &= -\cos x \\ f^{(4)}(x) &= \sin x \\ &\vdots \end{aligned}$$

Which gives

$$f(0) = 0,\ f^{(1)}(0) = 1,\ f^{(2)}(0) = 0,\ f^{(3)}(0) = -1,\ f^{(4)}(0) = 0, \ldots$$

So the Taylor expansion of order 4 is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}\cos c,$$

for some c (the interval containing c does not matter as we will see). The largest error possible in the approximation

$$\sin x \approx x - \frac{x^3}{6}$$

where $|x| < \frac{1}{2}$ is thus

$$\begin{aligned}\left|\frac{x^5}{5!}\cos c\right| &< \frac{1}{2^5 5!}\\ &= \frac{1}{32 \times 120}\\ &= \frac{1}{3840}.\end{aligned}$$

18. From previous question we have that the Taylor expansion of $\sin x$ about 0 on the interval $[0, x]$ is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + R_n(x),$$

where the remainder term $R_n(x)$ has absolute value

$$\left|\frac{x^{n+1}}{(n+1)!}\sin c\right| \quad \text{or} \quad \left|\frac{x^{n+1}}{(n+1)!}\cos c\right|,$$

depending on the value of n.
Now let $x = \frac{\pi}{5}$ and we have

$$\begin{aligned}|R_n(x)| &< \frac{1}{(n+1)!}\\ &< 0.00001,\end{aligned}$$

if $n = 8$. So we can use the approximation

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!},$$

to give

$$\sin\frac{\pi}{5} \approx 0.58778\ldots$$

Functions Exercises 3.4

1. a) $(u \circ v)(x) = x^2 + 2$
   b) $(v \circ w)(x) = \sin^2 x + 1$
   c) $(v \circ u \circ w)(x) = (\sin x + 1)^2 + 1$
2. $$f = R \circ 3\mathrm{Exp} \circ 2S$$
   and
   $$g = 2\mathrm{Exp} \circ S \circ (\mathrm{One} + R).$$
3. If we take the function f as $f : \mathbb{R} \to \mathbb{R}$ then f is one-to-one and onto and hence has an inverse. If we let $y = 3x + 1$ then $x = \frac{1}{3}(y - 1)$, so $f^{-1} : \mathbb{R} \to \mathbb{R}$ is defined by the formula
   $$f^{-1}(y) = \frac{1}{3}(y - 1).$$
   Sketch a graph of g. Note that g is undefined at $x = 3$. Taking g to be a function $g : \mathbb{R}\backslash\{3\} \to \mathbb{R}\backslash\{1\}$ then g is one-to-one and onto and so has an inverse. Setting $y = \frac{x}{x-3}$ we see that $x = \frac{3y}{y-1}$ so $g^{-1} : \mathbb{R}\backslash\{1\} \to \mathbb{R}\backslash\{3\}$ is defined by
   $$g^{-1}(y) = \frac{3y}{y - 1}.$$
   The function h is treated similarly to g. Taking h to be the function $h :: \mathbb{R}\backslash\{1\} \to \mathbb{R}\backslash\{1\}$ we see that h is one-to-one and onto and so has an inverse. Setting $y = \frac{x+1}{x-1}$ we see that $x = \frac{y+1}{y-1}$ and so $h^{-1} : \mathbb{R}\backslash\{1\} \to \mathbb{R}\backslash\{1\}$ is defined by
   $$h^{-1}(y) = \frac{y + 1}{y - 1},$$
   in other words $h = h^{-1}$.
4. Sketch a graph of λ. Setting $s = e^{\frac{t}{2}} - 1$ we see that $t = 2\ln(s + 1)$ and so $\lambda^{-1} : (-1, \infty) \to \mathbb{R}$ is defined by
   $$\lambda^{-1}(s) = 2\ln(s + 1).$$
5. First note that by the choice of domains and co-domains both *phi* and *psi* are one-to-one and onto and so both $\phi^{-1} : (-1, \infty) \to (0, \infty)$ and $\psi^{-1}(0, \infty) \to \mathbb{R}$ exist. They are defined by the formulas
   $$\phi^{-1}(x) = (x + 1)^{\frac{1}{2}}$$
   and
   $$\psi^{-1}(x) = \frac{1}{2}\ln x.$$
   Now $\phi \circ \psi : \mathbb{R} \to (-1, \infty)$ is defined by
   $$(\phi \circ \psi)(x) = \left(e^{2x}\right)^2 - 1 = e^{4x} - 1,$$
   and this is one-to-one and onto and so $(\phi \circ \psi)^{-1} : (-1, \infty) \to \mathbb{R}$ exists and is defined by the formula
   $$(\phi \circ \psi)^{-1}(x) = \frac{1}{4}\ln(x + 1).$$
   And checking $\psi^{-1} \circ \phi^{-1}$ we see that
   $$\begin{aligned}(\psi^{-1} \circ \phi^{-1})(x) &= \psi^{-1}(\phi^{-1}(x)) \\ &= \frac{1}{2}\ln\left((x + 1)^{\frac{1}{2}}\right) \\ &= \frac{1}{2}\frac{1}{2}\ln(x + 1) \\ &= \frac{1}{4}\ln(x + 1) \\ &= (\phi \circ \psi)^{-1}(x),\end{aligned}$$
   as required.

6. Not included.

Differentiation Exercises 4.6

1. Decide if the following limits exist and if so evaluate them.

a)
$$\lim_{x\to 3}\frac{2x-1}{3x+2}=\frac{5}{11},$$

b)
$$\lim_{x\to 2}\frac{3x-6}{2x}=0,$$

c)
$$\lim_{x\to 0}\frac{2\sinh x-2\sin x}{\cosh x-1}=0 \text{ apply l'Hopitals rule twice },$$

d)
$$\lim_{x\to 0}\frac{e^{x}+e^{-x}-2}{1-\cos(2x)}=\frac{1}{2} \text{ l'Hopitals rule again},$$

e)
$$\lim_{x\to 0}\frac{\sin(4x)}{\sin(6x)}=\frac{2}{3} \text{ l'Hopitals rule again},$$

f)
$$\lim_{x\to 0}\frac{x-\sin x}{x^3}=\frac{1}{6} \text{ l'Hopitals rule again},$$

g)
$$\lim_{x\to\infty}\frac{3x^3-6x}{3x+2x^3}=\frac{3}{2},$$

h)
$$\lim_{x\to 0}\frac{5x-6}{2x^3}, \text{ limit does not exist, function is unbounded},$$

i)
$$\lim_{x\to 0}(1+x)^{\frac{1}{x}}=e,$$

For this last one, let $y=(1+x)^{\frac{1}{x}}$, then $\ln y=\frac{\ln(1+x)}{x}$ and applying l'Hopitals rule to this we see that
$$\lim_{x\to 0}\ln y=1$$
and so
$$\lim_{x\to 0}y=e.$$
This last step is achieved by taking the exponential of both sides and using the fact that the exponential function is continuous.

2. Using the formal definition of continuity prove that if two functions f and g are both continuous at the point $x=a$ then so is the function $f+g$.

proof:

Let $\epsilon>0$. We need to find a $\delta>0$ so that the following statement is true
$$|x-a|<\delta\Rightarrow|(f+g)(x)-(f+g)(a)|<\epsilon.$$

Now we can manipulate the expression on the left hand side of the inequality to get

$$\begin{aligned} & |x-a|<\delta \Rightarrow |(f+g)(x)-(f+g)(a)|<\epsilon \\ \Leftrightarrow\ & |x-a|<\delta \Rightarrow |f(x)+g(x)-f(a)-g(a)|<\epsilon \\ \Leftrightarrow\ & |x-a|<\delta \Rightarrow |(f(x)-f(a))+(g(x)-g(a))|<\epsilon. \end{aligned}$$

But we can make use here of the inequality

$$\forall\, u,v \in \mathbb{R}\, |u+v| \leq |u|+|v|.$$

So

$$|(f(x)-f(a))+(g(x)-g(a))| \leq |(f(x)-f(a))|+|(g(x)-g(a))|.$$

But we know that f and g are both continuous at $x=a$ and so there exists $\delta_1, \delta_2 > 0$ satisfying

$$|x-a|<\delta_1 \Rightarrow |f(x)-f(a)|<\frac{\epsilon}{2},$$

and

$$|x-a|<\delta_2 \Rightarrow |g(x)-g(a)|<\frac{\epsilon}{2}.$$

Now to ensure that both of these inequalities will hold true we let $\delta = \mathrm{Min}(\delta_1, \delta_2)$. So now we can say that

$$|x-a|<\delta \Rightarrow |f(x)-f(a)|<\frac{\epsilon}{2},$$

and

$$|x-a|<\delta \Rightarrow |g(x)-g(a)|<\frac{\epsilon}{2}.$$

And adding the inequalities together we see that

$$|x-a|<\delta \Rightarrow |(f(x)-f(a))+(g(x)-g(a))| \leq |f(x)-f(a)|+|g(x)-g(a)| < \frac{\epsilon}{2}+\frac{\epsilon}{2}=\epsilon.$$

Clearing out the intermediate inequalities we see that this is equivalent to

$$|x-a|<\delta \Rightarrow |(f(x)-f(a))+(g(x)-g(a))|<\epsilon,$$

which is the statement we had to prove. This statement says that the function $f+g$ is continuous at the point $x=a$.

3. Prove from first principles that $\frac{d}{dx}(x^3)=3x^2$.

   Solution:

   Let $f(x)=x^3$ then

$$\begin{aligned} f'(a) &= \lim_{x\to a}\frac{f(x)-f(a)}{x-a}, \text{ the definition} \\ &= \lim_{x\to a}\frac{x^3-a^3}{x-a}, \text{ putting in the function } f \\ &= \lim_{x\to a}\frac{(x-a)(x^2+xa+a^2)}{x-a}, \text{ factorizing} \\ &= \lim_{x\to a}(x^2+xa+a^2)=3a^2, \text{ using algebra of limits theorem} \end{aligned}$$

   This argument holds for any $a \in \mathbb{R}$ and so the function $\frac{df}{dx}$ is given by $\frac{df}{dx}=3x^2$.

4. a) $6e^{2x}-10\sin(2x)$
   b) $2e^{4x}\,(2\cos(2x)-\sin(2x))$
   c) $e^{-x}x^{-2}\,(2x\cos(2x)-(x+1)\sin(2x))$
   d) $-3\sin(3x)e^{\cos(3x)}$

e) $8^x x^{3x}\left(3\ln(x)+3+\ln(8)\right)+(4x-7e^x)\cos\left(7e^x-2x^2\right)$

f) $\frac{-1}{\sin\left(\cos^{-1}(x)\right)}=\frac{-1}{\sqrt{1-x^2}}$

5. a) 2

   b) The tangent to the given curve at $(1,3)$ is vertical.

6. a) $\frac{dy}{dx}=\frac{4}{5}\coth(t)$

   b) $\frac{dy}{dx}=\frac{-1}{2}\frac{\sin(2t)}{\sin(t)}$

   c) $\frac{dy}{dx}=\frac{3t^2-4}{2t}$

7. $\frac{dy}{dx}=(x+3)^{x^2-4}\left(x^2+2(x+3)x\ln(x+3)-3\right)$

8. $$\frac{d}{dx}\left(\frac{uvw}{z}\right)=\frac{u'vwz+uv'wz+uvw'z-uvwz'}{z^2}$$

9. Let the rectangular part of the field have length l metres and width w metres. Then the perimeter of the track has length L given by

$$L=2l+2\pi w,$$

since $2l$ is the length of the two straight sides and $2\pi w$ is the perimeter of the two semi-circular ends. If the track is to have length 400m then

$$400=2l+2\pi w$$

and so

$$w=\frac{1}{2\pi}(400-2l)=\frac{1}{\pi}(200-l).$$

The area A of the rectangular part of the field is given simply by $A=lw$ and so

$$A=\frac{1}{\pi}(200l-l^2).$$

If this area is to be maximized then we look for a stationary point of A, i.e. a solution to

$$\frac{dA}{dl}=\frac{1}{\pi}(200-2l)=0.$$

This has one solution at $l=100$. Examining the second derivative shows that $\frac{d^2A}{dl^2}<0$ and so this is a local maximum, and in fact a global maximum.

So the area of the rectangular part of the field is maximized when $l=100$m.

10. Suppose that the cylindrical tin has dimensions of heigth h cm and radius r cm. Then the volume V of the tin is given by

$$V=\pi r^2 h=330\text{ cm}^3,$$

and so

$$h=\frac{330}{\pi r^2}.$$

Now the tin's surface area A is given by

$$A=2\pi r^2+2\pi rh,$$

where $2\pi r^2$ is the area of the two circular ends and $2\pi rh$ the area of the curved side of the tin. We can re-express A as

$$A=2\pi r^2+2\pi r\frac{330}{\pi r^2}=2\pi r^2+\frac{660}{r}.$$

We look for a stationary point of A by solving

$$\frac{dA}{dr} = 4\pi r - \frac{660}{r^2} = 0,$$

which is equivalent to

$$4\pi r^3 - 660 = 0$$

and so

$$r = \left(\frac{165}{\pi}\right)^{\frac{1}{3}} \approx 3.75.$$

Checking the second derivative shows that $\frac{d^2A}{dr^2} > 0$ at $r = 3.75$ and so this is indeed a local minumum.

So the tin's surface area is minimized when $r \approx 3.75$cm.